Electrical and Electronic Circuit

• Electrical and Electronic Difference + Defination?

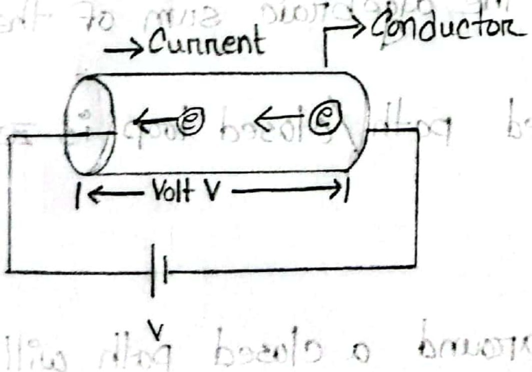
⇒ Electrical deals with the flow of electrical power or charge, where as electronics deals with the flow of electrons.

Example:-

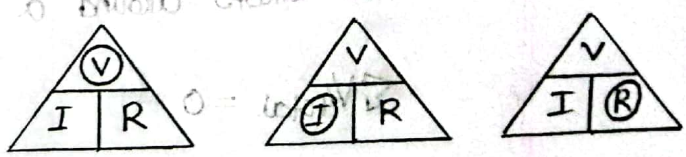
Electrical Devices → Transformers, generators, alternators, motors, circuit breakers, isolators.

Electronic Devices → Integrated circuit, logic gates, micro-processors, diodes, SCRs, transistors.

Ohm's Law:- Ohm's law state that the current throught a conductor between two points is directly proportional to the voltage across the two points, provide the temperature and other physical conditions remain unchanged.



Ohms Law Triangle



$V = IR$ $\quad I = \frac{V}{R}$ $\quad R = \frac{V}{I}$

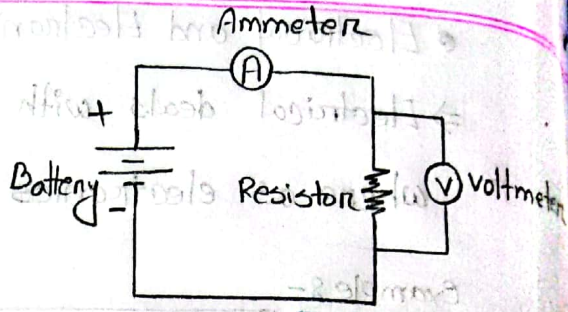
[P.T.O]

where,

V = Voltage

I = Current

R = Resistance



$V \propto I$

or, $\frac{V}{I} = \text{Constant} = R$

$\therefore V = IR$

Number of Cells	Current Throw the Wire (I (Ampere))	Potential Difference across the wire [V (voltage)]	R=
1	0.12	0.3	2.5
2	0.2	0.5	2.5
3	0.3	0.75	2.5

Kirchoff's Voltage Law (KVL):-

The algebraic sum of the potential rises and drops around a closed path / closed loop is zero.

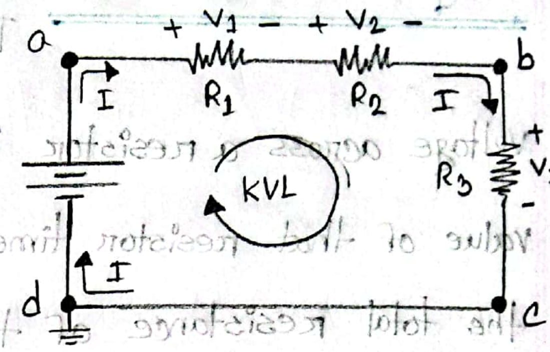
$$\sum V_{total} = 0$$

The sum of the voltage rises around a closed path will always equal the sum of the voltage drops.

$$\boxed{\sum V_{rise} = \sum V_{drops}}$$

$$-E+V_1+V_2+V_3=0$$

$$\therefore E = V_1+V_2+V_3$$



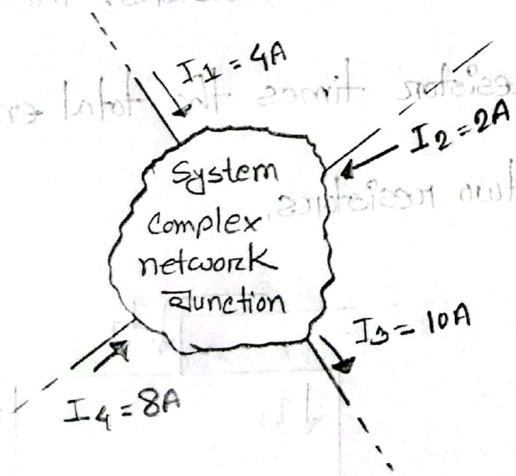
Kirchoff's Current Law (KCL):-

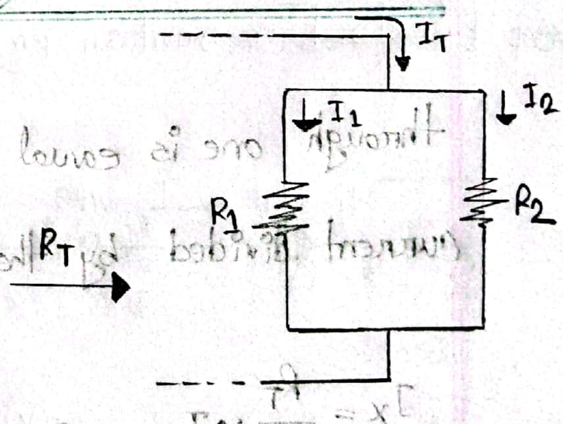
The algebraic sum of the currents entering and leaving a junction/region of network is zero

$$\Sigma I_{Total} = 0$$

The sum of the currents entering a junction of a network must equal the sum of the currents leaving the same junction.

$$\Sigma I_{entering} = \Sigma I_{leaving}$$





$$I_T = I_1 + I_2$$

$$\Sigma I_i = \Sigma I_o$$

or, $I_1 + I_4 = I_2 + I_3$

or, $4+8 = 2+10$

$\therefore 12A = 12A$

(checks)

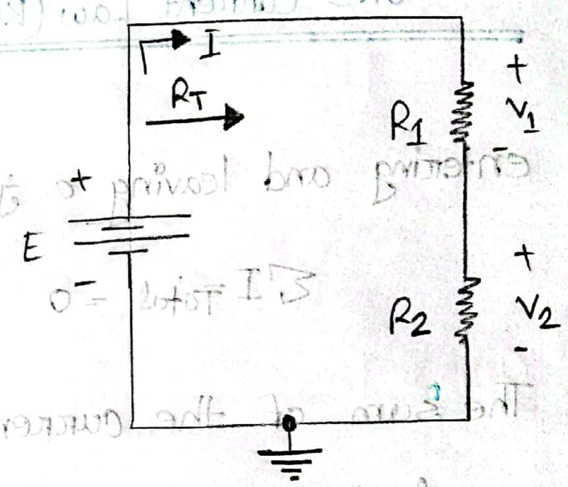
Voltage Divider Rule (VDR):-

The voltage divider rule states that the voltage across a resistor in a series circuit is equal to the value of that resistor times the total applied voltage divided by the total resistance of the series configuration.

$$V_x = \frac{R_x}{R_T} \times E$$

$$\Rightarrow V_1 = \frac{R_1}{R_1 + R_2} \times E$$

$$\Rightarrow V_2 = \frac{R_2}{R_1 + R_2} \times E$$



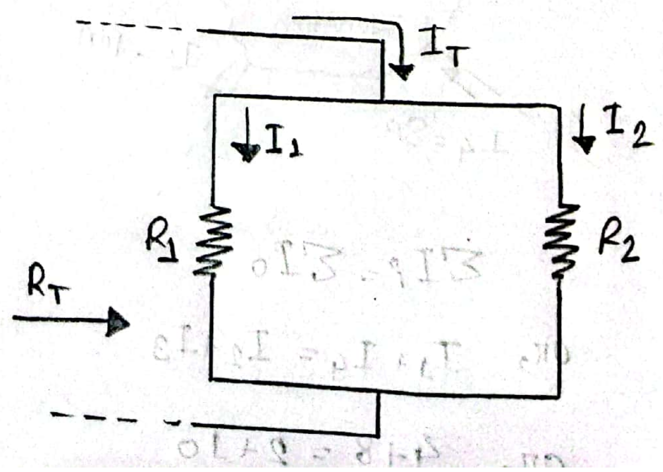
Current divider rule (CDR):-

For two parallel resistors, the current through one is equal to the other resistor times the total entering current divided by the sum of the two resistors.

$$I_x = \frac{R_T}{R_x} \times I_T \quad \text{—— (i)}$$

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\Rightarrow \frac{1}{R_T} = \frac{R_1 + R_2}{R_1 R_2}$$



$$\Rightarrow R_T = \frac{R_1 R_2}{R_1 + R_2}$$

$$I_1 = \frac{\left(\frac{R_1 R_2}{R_1 + R_2}\right) \times I}{R_1}$$

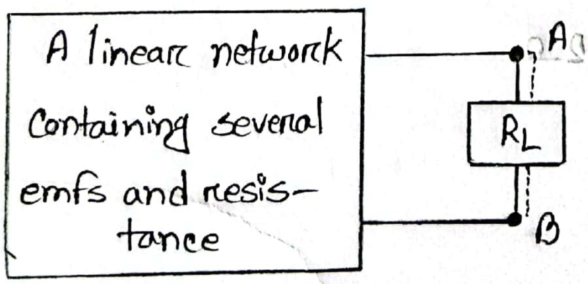
$$\Rightarrow I_1 = \frac{\cancel{R_1} R_2 \times I}{(R_1 + R_2)\cancel{R_1}}$$

$$\Rightarrow I_1 = \frac{I R_2}{R_1 + R_2}$$

Same, $I_2 = \dfrac{I R_1}{R_1 + R_2}$

Thevenin's Theorem :-

Thevenin's theorem states that "Any linear circuit containing several voltage and resistance can be replaced by just one single voltage in series with a single resistance connected across the load".



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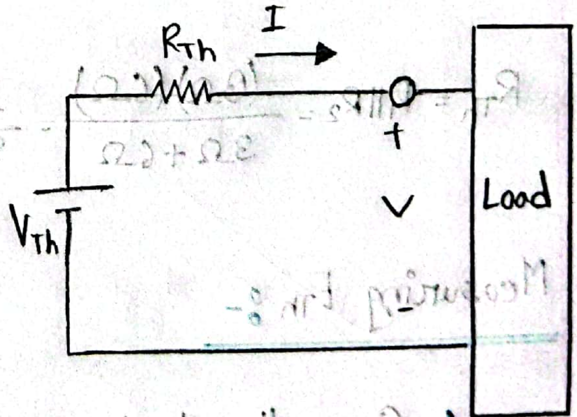
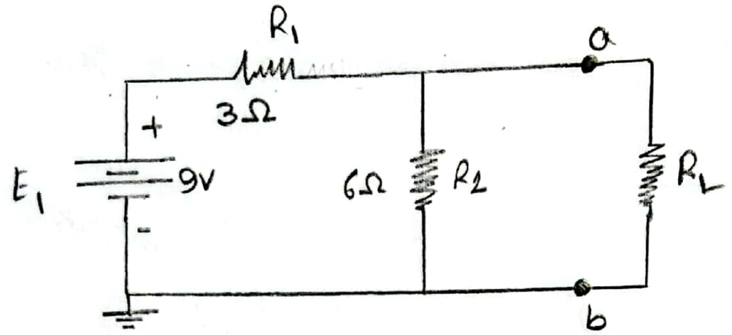


Measuring R_{Th} :-

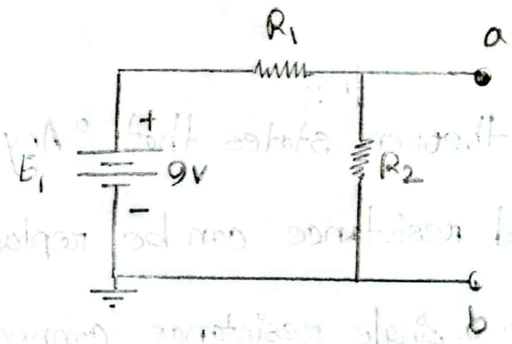
1. Open the load resistor.

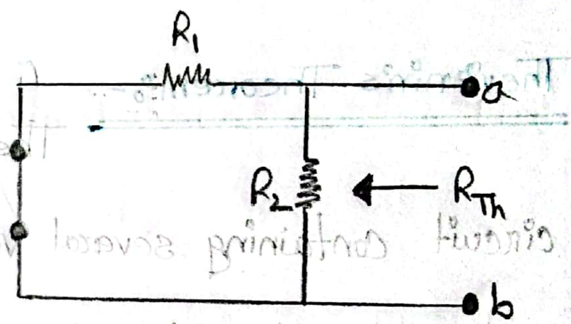
2. Open current sources and short voltage sources.

3. Calculate/measure the open circuit resistance. This is the Thevenin Resistance (R_{Th}).



R_{Th} Example :-





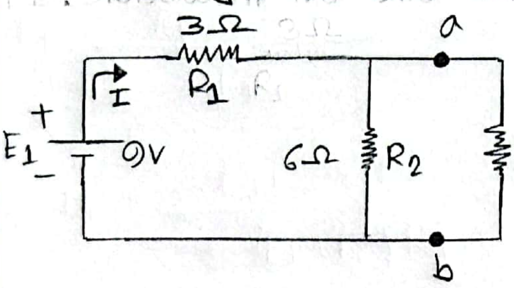
$$R_{Th} = R_1 \| R_2 = \frac{(3\Omega)(6\Omega)}{3\Omega + 6\Omega} = \frac{18}{9} = 2\Omega$$

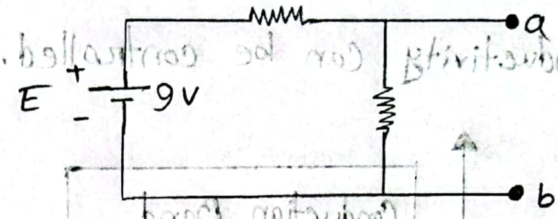
Measuring E_{Th} :-

1. Open the load resistor.

2. Redraw the circuit with voltage source and current source.

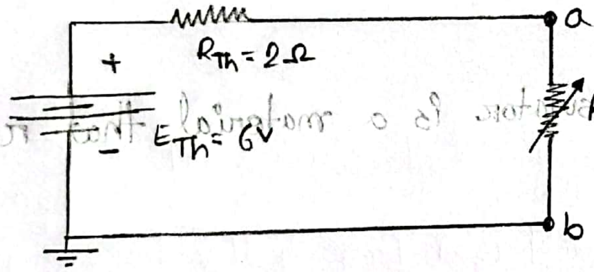
3. Calculate/measure the open circuit voltage. This is the Thevenin Voltage.





$$E_{Th} = \frac{R_2 E_1}{R_2+R_1} = \frac{6\times 9}{6+3} = \frac{54V}{9} = 6V$$

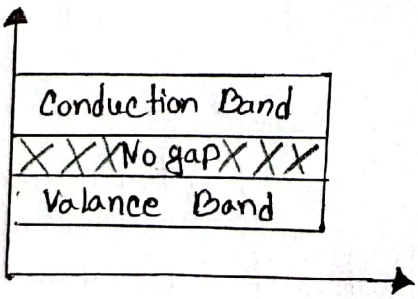
Thevenin equivalent circuit:-



$$I_L = \frac{V_{Th}}{R_{Th}+R_L} \qquad V_L = I_L \times R_L$$

Conductors:-

Conductor is a material that allows the flow of electric charge with minimal resistance.

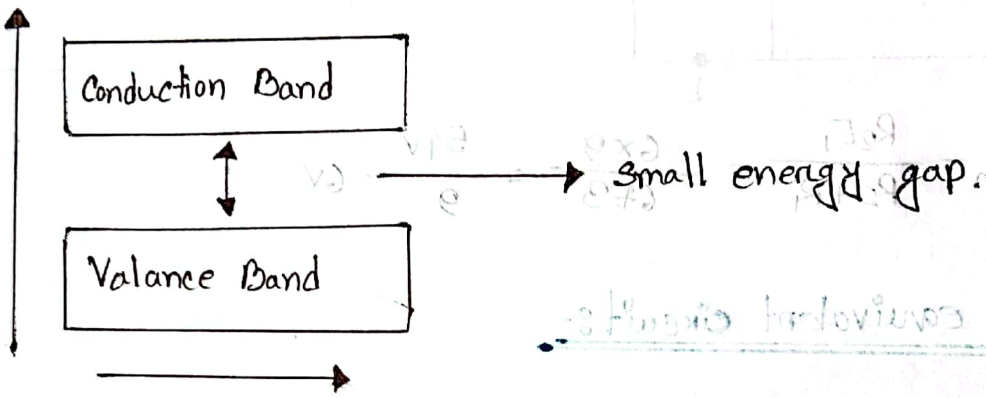


Energy Band

in conductor, energy band and Valance band overlaps.

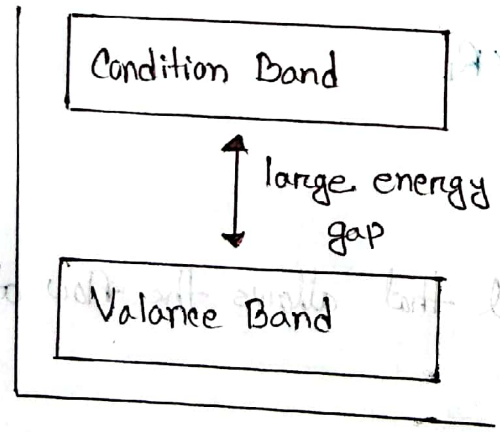
Semi-Conductor :-

Semi-conductor is a material with electric conductivity between that of a conductor and an insulator. It's conductivity can be controlled.



Insulator :-

An insulator is a material that rasists the flow of electrical current.



in insulator conduction band and valance are separated by large energy gap.

Energy band diagram

Difference :- Conductor, Semi-conductor & Insulator,

	Conductor	Semi-conductor	Insulator
Conductivity	High	Moderate	Law
Band gap	No band gap	Small band gap	Large band gap
Electrical Resistivity	Very low	Moderate	High
Electron flow	Free movement of electrons	Limited	No free electrons under normal condition.
Example	Metals, Copper, Silver	Silicon, Gallium, Arsenice	Rubber, Glass, Plastic.